# Dissertation Proposal

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# Statement of the Problem

**General Perspectives**

🏍️ Motorcyclists account for ~21% of global road deaths (WHO, 2023).

🛣️ At signalized junctions, riders filter between stopped vehicles, swerving, overtaking, lane shifting, queue jumping, which create complex interactions.

🔄 Current models poorly capture these behaviors and their safety risks.

🚁 This study uses drone trajectories and near-miss indicators to improve junction safety.

**Academic Perspectives**

↔**Gap 1:** Lane-filtering and other motorcycle maneuvering behaviors lack data-driven criteria.

📊 **Gap 2:** Motorcycle-specific safety thresholds and composite indices are not available.

🔍 **Gap 3:** Limited empirical validation of SUMO's sub-lane model for motorcycle lane-filtering.

🎯 **Research Aim:** Develop an integrated framework to characterize motorcycle maneuvering, assess safety, and simulate motorcycle behavior.

# Scope of the Study

📍 **Setting:** Signalized intersection approaches with mixed traffic (motorcycles, cars, large vehicles).

🎥 **Data:** Drone video, signal timing, and road geometry. Georeferenced trajectories are extracted and validated.

🛵 **Focus:** Motorcycle–vehicle interactions and lane filtering, including lateral shifts, overtaking, swerving, and queue jumping.

⚠️ **Safety Measures:** TTC, PET, DRAC, and lateral clearance; elliptical safety envelope for deriving safety thresholds and a composite safety index.

🧠 **Modelling:** Multinomial logit model for maneuver choice and SUMO-based hybrid motorcycle behavior model, validated using independent trajectories and benchmarked against the default sub-lane model.

# Limitations of the Research

🕒 **Observation period:** Drone footage covers limited times and conditions, so peak and off-peak, weather and lighting variation may not be fully captured.

⚠️ **Surrogate measures, not crashes:** TTC, PET, DRAC and the safety index indicate risk but are not crash probabilities. Thresholds are percentile-based, so they show *relative* severity.

🧠 **Behavioral scope:** Rider intent, experience and attention cannot be observed from video, so the models describe behavior but not its causes.

🔍 **Manual validation:** Video review is done on a subset only and involves some subjectivity.

⚙️ **Site coverage:** Findings come from a limited number of signalized approaches, so results may not transfer to other geometries, signal plans or traffic compositions.

💻 **Software dependence:** The model is built for SUMO, so it may not carry over directly to other simulators.

# Research Landscape

**Past: Historical Development**

🚗 Early safety work counted crashes, then moved to near-miss measures like TTC, PET and DRAC (TTC, PET, DRAC) (Maji & Ghosh, 2025; Samalla & Chunchu, 2024; Zheng & Sayed, 2019)

🏍️ Filtering riders' lateral clearance was measured with on-board sensors and regression (Promraksa et al., 2022).

🔄 Driving safety field: risk from driver–vehicle–road interaction (Avathkattil & Vedagiri, 2026; Wang et al., 2015).

🛣️ SUMO sub-lane model: lanes divided into lateral strips (Eclipse SUMO, n.d.; Minh et al., 2012; Lee et al., 2009)

✅ **Strength:** simple and interpretable.

⚠️ **Weakness:** car-centered and mostly longitudinal.

# Research Landscape (Contd.)

**Cutting Edge**

- 🎯 Two-dimensional indicator (Anticipated Collision Time) (Avathkattil & Vedagiri, 2026; Samalla & Chunchu, 2024)
- 🛵 Evasive-action indicators for powered two-wheelers (Ezzati Amini et al., 2022; Phan et al., 2016; Lee et al., 2009)
- 🔀 Prediction of motorcycle filtering choice (Damani & Vedagiri, 2023; Promraksa et al., 2022)
- 🚁 Composite conflict indicator from drone trajectories (Jain et al., 2025); Maji & Ghosh, 2025; Wu et al., 2020)
- 🛣️ Sublane model validated for heterogeneous traffic (Wichitphongsa et al., 2025; Minh et al., 2012; Lee et al.,2009)

📊 **Strength:** high-resolution trajectory data.

📍 **Weakness:** limited sites and validation.

# Research Landscape (Contd.)

🔍 **Gap**

🏍️ No calibrated motorcycle safety envelope.

🔗 2D interaction, thresholds, composite safety index, and SUMO model are not integrated.

🎯 **This study:** Safety envelope, 2D safety index, and simulation in one validated framework.